Database Lab : Relational Join Strategy & Plan Analysis Setup

Duration : 30 Min.

Scenario

As a Query Optimization Engineer at NIIT, an enterprise reporting query linking departments and employees requires performance tuning. You must create two relational tables with primary/foreign keys in PostgreSQL, populate sample records, and record the optimizer's join strategy in an audit table.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!` | CLI: `psql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **Departments Table:** `<prefix>_depts` - **Employees Table:** `<prefix>_emps` - **Join Audit Table:** `<prefix>_join_strategy_audit`

Task Objectives

Perform the following configuration steps directly via `psql`:

1. Create Relational Tables with Keys

- Create `<prefix>_depts (dept_id INT PRIMARY KEY, dept_name VARCHAR(50) NOT NULL)`.
- Create `<prefix>_emps (emp_id SERIAL PRIMARY KEY, dept_id INT REFERENCES <prefix>_depts(dept_id), emp_name VARCHAR(50) NOT NULL, salary NUMERIC(10,2))`.
- Insert 3 departments and 10 employee rows.

2. Create Join Strategy Audit Table

- Create table `<prefix>_join_strategy_audit` with columns:
- `id SERIAL PRIMARY KEY`
- `join_type VARCHAR(30) NOT NULL`
- `operator_name VARCHAR(30) NOT NULL`
- `tested_by VARCHAR(50) NOT NULL`

3. Record Plan Operator Verification

- Insert a record logging `join_type = 'INNER JOIN'`, `operator_name = 'Hash Join'`, and `tested_by = '<prefix>'`.

Verification

Once completed, run the verification script:

```
python run.py
```

The driver connects to PostgreSQL, validates the foreign key relationships, and checks the join audit table.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Primary & Foreign Key Tables (_depts, _emps) Creation	3 Marks
TC2	Data Seeding & Referential Integrity Verification	3 Marks
TC3	Join Audit Table (_join_strategy_audit) Setup	2 Marks
TC4	Join Operator (Hash Join vs Nested Loop) Verification	2 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- Foreign key constraint must be active.